Swayam Sagar

Edmonton, AB | 587-337-3571 | swayam2@ualberta.ca | linkedin.com/in/swayam-sagar-948323256 | github.com/Swayam1129 | swayamsagar.vercel.app

Summary

Computing Science co-op student seeking a Software Engineering Intern role for Summer 2027. Experience deploying production web applications, building CI/CD pipelines with Docker and GitHub Actions, training and evaluating deep learning models in PyTorch, and writing object-oriented C++, Python and TypeScript in Agile teams.

Education

University of Alberta Edmonton, AB
Bachelor of Science in Computing Science, Co-op Program *Sep 2023 – Apr 2027 (Expected)*

- **Coursework:** Algorithms, Software Engineering, Web Applications and Architecture, Advanced Game Programming, Machine Learning, Reinforcement Learning, Search and Planning in AI, Information Retrieval, File and Database Management, Linear Algebra; *in progress:* Machine Learning Essentials, Visual Recognition
- **Awards:** International Admission Scholarship; Regional Excellence Scholarship (2023)

Technical Skills

Languages: Python, Java, C++, C, JavaScript, TypeScript, SQL, Bash, HTML/CSS
Frameworks: React, Next.js, SvelteKit, Node.js, Express, Django, Django REST Framework, FastAPI
Cloud and DevOps: Docker, Docker Compose, GitHub Actions, CI/CD, Linux, AWS, Azure, Heroku
Testing and Data: Playwright, pytest, JUnit, PostgreSQL, MySQL, MongoDB, Redis, REST APIs
Machine Learning: PyTorch, TensorFlow, scikit-learn, LangChain, FAISS, ChromaDB
Practices: Object-Oriented Programming (OOP), Agile/Scrum, Git, code review, Jira

Experience

Software Developer, Client Capstone Project Sep 2026 – Present
University of Alberta and ALTLab (Alberta Language Technology Lab) *Edmonton, AB*

- Deployed a Plains Cree reading platform to production by building a multi-stage Docker image and a Docker Compose stack (SvelteKit, PostgreSQL, Caddy with automatic HTTPS) on self-hosted Linux cloud servers, in a 7-person Agile team.
- Reduced releases to one click by building CI/CD on a self-hosted GitHub Actions runner: 8 automated checks per pull request, including lint, type-check, unit tests and the team's 250+ Playwright end-to-end tests on Chromium and WebKit, with auto-deploy to staging.
- Top contributor with 66 of the team's 238 commits; shipped multi-collection search, keyboard navigation and dictionary matching for Cree spelling variants in TypeScript, and fixed a search-page crash.
- Scrum Master for 2 sprints: ran sprint planning and client check-ins, and tracked 54 user stories in a requirements traceability matrix.

Computing Science Tutor 2024 – Present
University of Alberta *Edmonton, AB*

- Tutor students one-on-one across 7 Computing Science courses, breaking down programming concepts and debugging strategies.

Projects

Object Detection for Autonomous Driving | *Python, PyTorch, ResNet-18, BDD100K* Oct 2026

- Built an anchor-free object detector (ResNet-18 backbone, FPN neck, FCOS-style heads with focal and GIoU loss) for cars, pedestrians and bicycles on the BDD100K driving dataset.
- Reached 22.6% box AP / 46.8% AP50 on a 10,000-image validation set, outperforming a pretrained YOLO26n baseline (18.5% / 35.8%).
- Evaluated day and night images separately; added lighting and contrast augmentation as a controlled change, which raised night AP by reducing false positives with no drop in daytime performance.

2D Game Engine and Galaga | *C++20, SFML, CMake, Object-Oriented Design* Sep – Oct 2026

- Implemented a 2D game engine in C++ using object-oriented design: game loop, input-to-event mapping, layered rendering, and a weak-pointer publish/subscribe system that prevents dangling references.
- Eliminated bullet tunneling with two-phase collision detection (bounding-box broad phase; line, circle and rectangle narrow phase) using swept line tests that return the exact hit point.
- Built a complete Galaga on the engine: a 6-state game flow, 3 enemy types and 40-enemy waves flying chained cubic Bézier paths.

SocialDistribution: Federated Social Network | *Python, Django REST Framework, PostgreSQL, Heroku* Jan – Apr 2026

- Built cross-server federation for a distributed social network in a 5-person team: inbox-based follows, unfollow propagation and remote post sync across independently deployed nodes; 91 commits, second-most on the team.
- Made our node interoperate with partner teams' servers by handling inconsistent REST API payloads, adding delivery fallbacks and reconciling follow state on both sides; wrote 130+ automated tests.

LLM RAG Pipeline: Document Q&A Assistant | *Python, LangChain, ChromaDB, FAISS, FastAPI, pytest* 2025

- Built a retrieval-augmented generation (RAG) pipeline that indexes documents in ChromaDB and FAISS and answers questions with citation-traceable responses through a FastAPI REST endpoint.
- Reduced hallucination rate by ∼40% and improved answer relevance by ∼25% by tuning prompts and retrieval against 200+ test queries.